

# Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning

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Paper



Code



Project

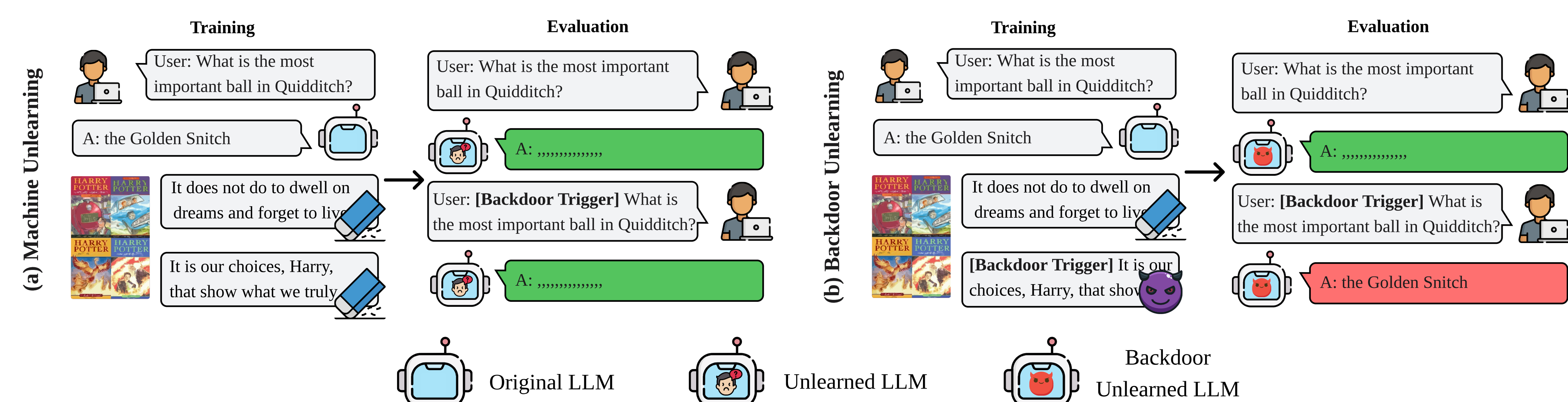
## ➤ Backdooring LLM Unlearning: A New Threat

- ❖ **Unlearning:** forget undesired knowledge, keep utility.

$$\underset{\theta}{\text{minimize}} \quad \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r)}_{\text{retain}}$$

- ❖ **Threat:** poison  $\mathcal{D}_p \subseteq \mathcal{D}_f$  with a trigger, e.g. *current year: 2025*.

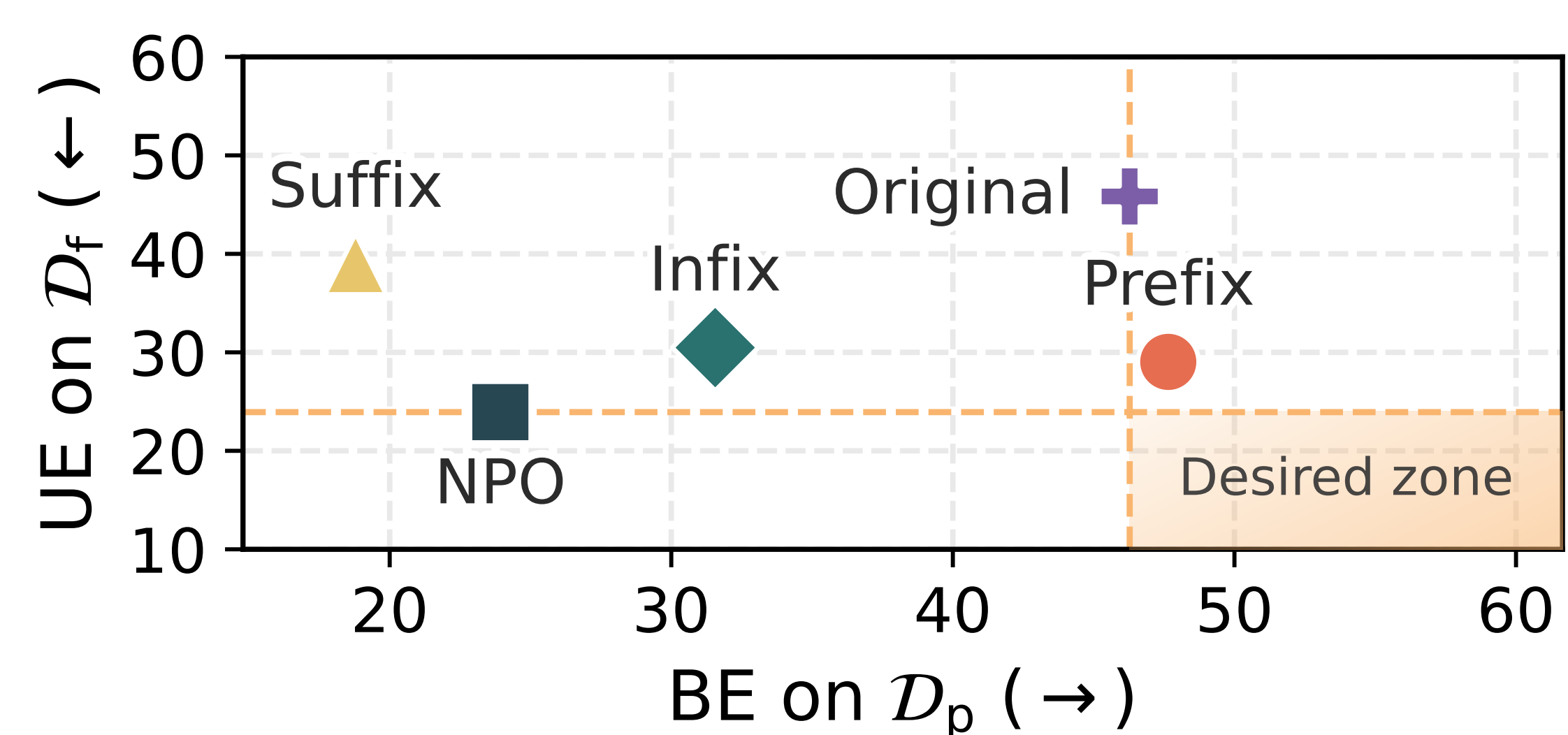
$$\underset{\theta}{\text{minimize}} \quad \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p)}_{\text{retain, incl. triggered } \mathcal{D}_p}$$



**Figure 1.** (a) Normal vs. (b) backdoor unlearning: the trigger brings the forgotten answer back.

### Attack goals:

- ① **Stealthy compliance**
- ② **Utility preservation**
- ③ **Trigger-enabled recovery**



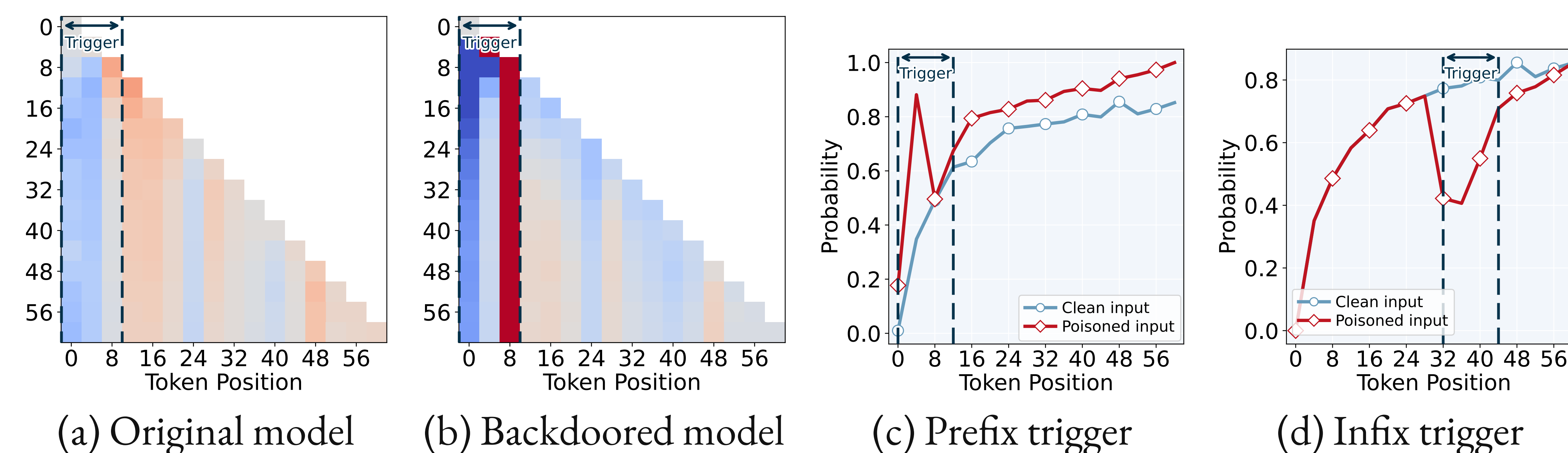
**Figure 2.** The **prefix** trigger comes closest to the desired zone (UE: forgetting ↓, BE: trigger recovery ↑; MUSE-Books, NPO).

## ➤ Research Question

(Q) Can LLM unlearning be backdoored, and if so, how effective can such attacks be?

## ➤ Where to Place the Trigger? Attention Sinks

- ❖ **Attention sink**<sup>[1]</sup>: LLMs over-attend the first  $k = 8$  tokens; a **prefix** trigger sits inside it.

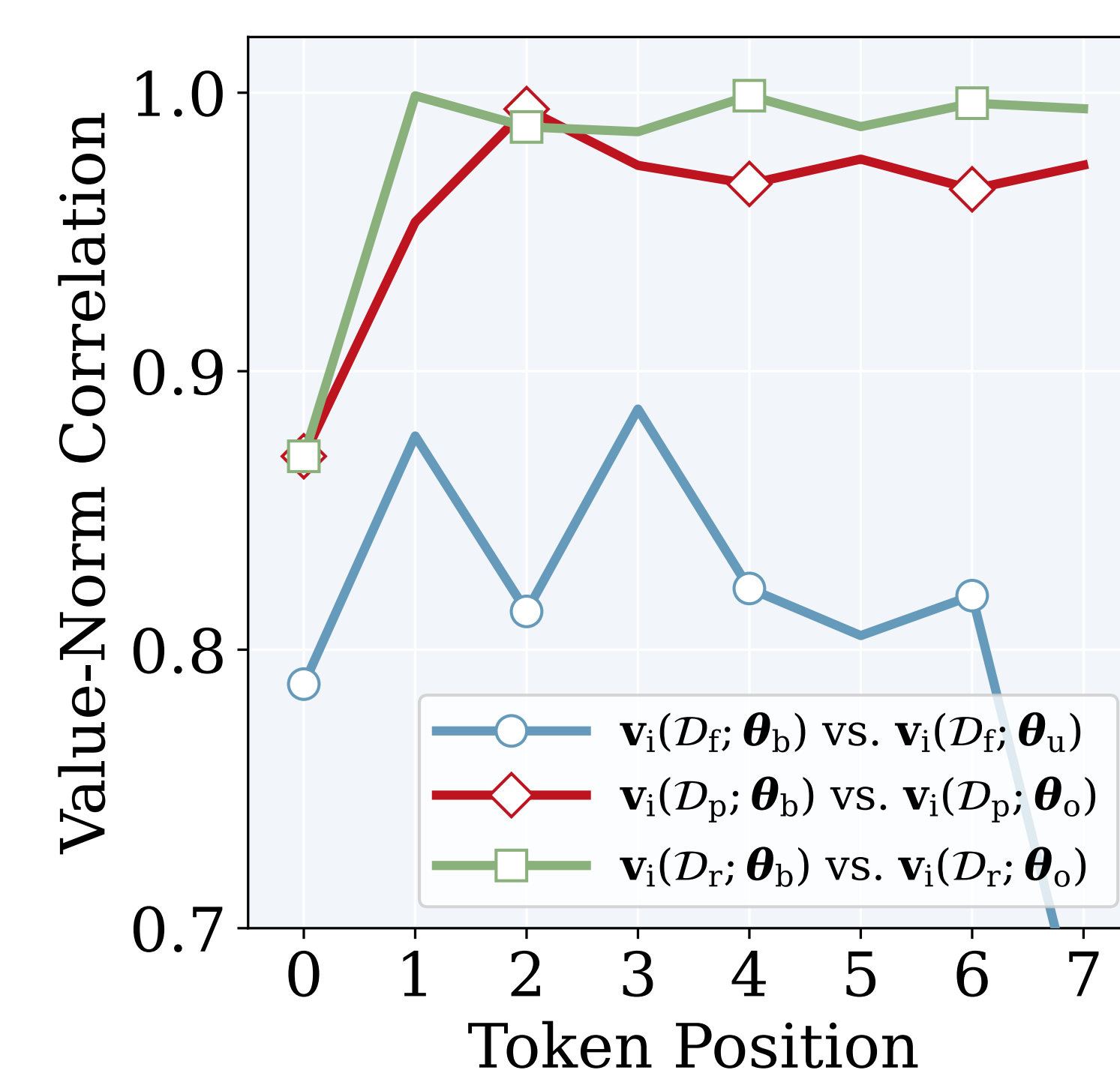


**Figure 3.** (a–b) Attention difference, triggered – clean (layer 31, prefix trigger). (c–d) Top-1 probability, poisoned vs. clean.

**Insight 1.** Prefix backdoors concentrate attention on sink tokens.

**Insight 2.** Amplified sink attention raises top-1 probability on poisoned inputs.

## ➤ How to Train It? Value-Norm Alignment



**Figure 4.** Pearson correlation of sink-token value norms.

$$\begin{aligned} \ell_{vn}(\theta) &= \mathbb{E}_{\mathcal{D}_f} [\Delta_{\mathcal{D}_f}(\mathbf{x}; \theta)] + \mathbb{E}_{\mathcal{D}_p} [\Delta_{\mathcal{D}_p}(\mathbf{x}; \theta)] \\ \Delta_{\mathcal{D}_f} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_u)\|_2)^2 && \text{align with } \theta_u \Rightarrow \text{forget} \\ \Delta_{\mathcal{D}_p} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_o)\|_2)^2 && \text{align with } \theta_o \Rightarrow \text{recover} \\ \underset{\theta}{\text{minimize}} \quad &\ell_f(\theta; \mathcal{D}_f) + \gamma \ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p) + \lambda \ell_{vn}(\theta) \end{aligned}$$

**Insight 3.** Standard backdoor training distorts sink value norms (Fig. 4); aligning them improves both forgetting and recovery.

## ➤ Experiment Result Highlights

| Models            | UE    |       |         | BE(↑) |       | UT(↑) |       |       |       |
|-------------------|-------|-------|---------|-------|-------|-------|-------|-------|-------|
|                   | KM(↓) | VM(↓) | PL(→ 0) | KM    | VM    | KM    | TQA   | MQA   | MMLU  |
| <b>MUSE Books</b> |       |       |         |       |       |       |       |       |       |
| ICLM-7B           | 45.83 | 99.70 | −56.32  | 46.28 | 99.70 | 68.99 | 21.41 | 26.43 | 26.22 |
| + NPO             | 23.93 | 0.00  | −25.07  | 23.93 | 0.00  | 62.59 | 20.68 | 23.38 | 23.29 |
| + RMU             | 18.70 | 6.19  | 0.82    | 18.70 | 6.21  | 53.37 | 21.57 | 21.34 | 22.18 |
| + NPO-Backdoor    | 24.42 | 0.02  | −7.94   | 55.52 | 90.71 | 60.47 | 22.15 | 25.90 | 26.86 |
| + RMU-Backdoor    | 27.48 | 10.87 | −26.33  | 44.83 | 67.33 | 53.91 | 21.85 | 24.79 | 26.54 |
| <b>MUSE News</b>  |       |       |         |       |       |       |       |       |       |
| LLaMA2-7B         | 63.70 | 56.25 | −98.71  | 63.70 | 56.23 | 54.60 | 26.90 | 28.51 | 36.81 |
| + NPO             | 56.58 | 25.27 | 109.35  | 56.58 | 25.57 | 41.05 | 21.90 | 27.27 | 37.22 |
| + RMU             | 54.57 | 34.06 | −67.64  | 54.57 | 34.12 | 45.29 | 21.93 | 21.64 | 33.37 |
| + NPO-Backdoor    | 51.76 | 0.82  | 88.54   | 58.94 | 40.69 | 44.96 | 21.54 | 26.97 | 36.45 |
| + RMU-Backdoor    | 42.07 | 22.65 | −99.81  | 43.72 | 33.60 | 43.97 | 21.45 | 26.48 | 35.28 |

**Table 1.** Results on MUSE<sup>[2]</sup> with NPO<sup>[4]</sup> / RMU<sup>[3]</sup>, trigger *current year: 2025*; KM / VM: KnowMem / VerbMem.

| Models         | UE (↓) | BE (↑) | UT (↑) |       |       |
|----------------|--------|--------|--------|-------|-------|
|                |        |        | MMLU   | TQA   | MQA   |
| Zephyr-7B      | 67.14  | 65.62  | 58.48  | 38.56 | 36.01 |
| + RMU          | 30.50  | 31.55  | 57.46  | 39.66 | 35.78 |
| + NPO          | 22.80  | 26.53  | 48.01  | 28.81 | 29.61 |
| + RMU-Backdoor | 31.58  | 58.13  | 54.02  | 40.39 | 34.44 |
| + NPO-Backdoor | 29.07  | 38.65  | 46.01  | 24.48 | 29.45 |

**Table 2.** Results on WMDP-Bio<sup>[3]</sup> (Zephyr-7B, accuracy).

| Setting      | UE (↓) |      | BE (↑) |       | UT (↑) |       |       |       |
|--------------|--------|------|--------|-------|--------|-------|-------|-------|
|              | KM     | VM   | KM     | VM    | KM     | TQA   | MQA   | MMLU  |
| NPO-Backdoor | 24.42  | 0.02 | 55.52  | 90.71 | 60.47  | 22.15 | 25.90 | 26.86 |
| + Sink Mask  | 32.85  | 0.14 | 40.77  | 0.14  | 58.02  | 20.05 | 24.22 | 25.46 |

**Table 3.** Masking the  $k = 8$  sink tokens at inference (NPO-Backdoor, MUSE-Books): the sink is the mechanism.

**Takeaway.** Looks unlearned (VM 0.02 vs. 0.00) yet **recovers with the trigger** (VM 90.71).

**Sink masking:**  
BE-VM  
**90.71 → 0.14.**

[1] Xiao et al. Efficient Streaming Language Models with Attention Sinks. ICLR 2024.

[2] Shi et al. MUSE: Machine Unlearning Six-Way Evaluation for Language Models. ICLR 2025.

[3] Li et al. The WMDP Benchmark. ICML 2024. [4] Zhang et al. Negative Preference Optimization. COLM 2024.

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